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Hermit Crabs Evolve Employment Pattern of Human

Abstract: Today's human resources is a challenge faced by the flow of talent problem, the flow of talent includes two aspects, one is the flow of talent between organizations, commonly referred to as brain drain; Second, the flow of talent within the organization. At present, the theory and practice of community pay more attention to the first problems, and they have accumulated a wealth of theoretical and empirical. There has been relatively mature approach about how to deal with the problem of brain drain. In contrast, the flow of talent within the organization is lack of appropriate research. When organizations face such problem, they often use existing experience and sometimes helpless.

By observing the hermit crab shedding model, we be inspired: if we set the appropriate exchange mode, you can make more job seekers get a chance to exchange works; In this mode, we define a comprehensive indicator of the ability to work seekers, sort job seekers according to it, let them wait to right jobs appear, then exchange works successively; however, there is a severe shortage, such pattern doesn't consider the current situation of job seekers and the requirement of firms. Thus we propose to improve the model.

Via a collaborative approach and restrictions, improved model can make the exchange more in line with the actual situation, and also speed up the exchange of work. Collaborative approach is: We calculate what companies need new staff, and then we provide suitable job seekers to them. When current income of the seeker is below the new work and job seeker's ability is greater than the requirement of the new work, exchange happens.

Through simulation experiments, we found that different mobile chain lengths will impact the fluctuation of the flow chain. In general, with the increase of the chain length, the applicant's average exchange benefit will be gradually reduced, but overall yields will increase. They are basically linear. At the same time, if randomly assigned a chain length, the fluctuation of crossing-over value is too much, which is not conducive to management of human resources. So, we set the cycle be 7 days, the flow chain be 3-133 in steps of 10 and 15-35 in steps of 2 simulate respectively. By analyzing the changes of the mean of crossing-over value, in that of the variance and the coefficient of variance, we found that the best length of the flow chain is about 20, which is the assumed number of new posts every week.

Finally, we analyze the changes of human resource in 5 years with the chain length of 19, we found that the mean of the benefit of every exchange, mainly lies in $[0.07, 0.09]$, and the total of the benefit within every week mainly lies in $[1.2, 1.6]$, which means the fairness our model is better than tradition.

Key words: AHP, Collaborative Model, Opportunity Chain, Human Resource

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I. Introduction

1.1 The background of changing jobs

In the current society, employment conditions is very serious. The Labor market is now facing the huge pressure that more than 1400 employees laid off from state-owned enterprises, 150 million rural surplus labor, and nearly 10 million new Labors from towns each year. According to forecast of authority, the next 3 to 5 years, the town needs to arrange 22 ~ 22 million people to work each year, even if the economic growth rate stood at 8%, it can still

appear tens of millions employment gap. The employment pressure from university graduates, the second-generation migrant workers and low-income people, is more outstanding. Facing huge employment market in China, it's great challenge of opportunity for individuals and companies. The work package assigned time has passed. The two-way channel between the enterprise and job seekers makes enterprise and job seekers have many choices. Finding the right job and recruiting elite talent, have become their primary goals. But, due to the problem of way, the staff is not suitable for this position and "incompetent", or the talent in the position is too good and "overqualified", which will let individuals and enterprises have a big loss finally.

A lot of people inevitably come up to changing jobs when they have begun working for a period of time. A survey participated by over 6500 white-collar workers, which is made by a domestic professional human resource service today shows that 20.8% of white-collar workers having already completed the job-hopping, 32.2% having begun to find a new job opportunities, and 35.4% having inclination, but no action, 11.6% not trying to jump ship. As a office worker, do have a lot of bitterness, cut, demotion, impotence and so on. Many people secretly hope to change job. This ratio is particularly higher among young people. By comparison, job-hopping intention of Post-90's generation is higher than predecessors, reaching 67%. At the same time, enterprises under the market economy pay more attention to benefit, and also eager to adjust resource[1].

In general, The main reason why people try to change their jobs are the family and getting a higher salary and better career development opportunities, for a change of environment, the identification of values, relieving pressure, and learning new skills and creative thinking. Changing jobs is a good thing, which can prove his confidence and ability of scholars in the workplace. Many successful workers are now not engaged in his first job, or even have nothing to do with the first job. But we should see that job-hopping is, after all, the risk of a "betting". The question of changing jobs now not seriously but more and more following one's inclinations, which creates the following questions:

- Not favorable to the development of the company. Employees are like "parts" on the machine, job-hopping of the employee will lead to the company operation efficiency to reduce and leak trade secrets. The new staffs also need to get up to

speed. The cycle of training skilled workers and the cost of training, are not favorable to the development of the company.

- Employment pressure to the society. Social employment pressure is opposite bigger nowadays, meaningless job-hopping will only increase the employment difficulty, ignore the social stability.
- Mercenary. To make money, is everyone's direct job purpose, otherwise, taking the pursuit of higher income as the only goal of the transformation of work, ignoring their own conditions and the need of further development, at the expense of their own long-term interests for immediate temporary interests, also is not worth the cost.
- Go with the flow. In seeing a certain industry is very hot, we choose this kind of work, just go with the flow, the gaps. Frequently careers-hopping are often the result of not learning the new technology, but abandoning the original knowledge, skills.
- Unbalance. They haven't weighed their abilities carefully before changing jobs, haven't seen if their abilities can reach the level to change a better job. They shift work easily even their ability can not meet the required level by the new job, which will result in not getting the new job and losing the original job.

We will analyze the main reason why the problems occur from the following aspects. For individuals, lack of self objective understanding and evaluation; without developing a sense of urgency, which leads to the absence of career planning consciousness; unclear professional position, no career planning; only a goal, but no feasible solution and the path of development and professional mentality is not mature, when frustrated easily give up, which are why he can not get ideal jobs, and not take the biggest use of their talents to the extreme. For the company: the lack of corporate culture which makes employees not to produce a sense of belonging, and the imperfect system which will make people question the future of their career and low wages which will let a person feel the pressure of life. It will lead to that it's hard for the company to retain the backbone talents, and cannot let talents be on the most appropriate post taking use of their talents to the extreme and create the biggest value for the company.

1.2 The traditional way of applying for a job

- The job fair at the scene. Job fairs have two categories, one is a comprehensive job fairs. This kind of job fairs are a lot, but the suitable companies may not many. Another kind is industry-wide special job fairs, this kind of job fairs is few, only twice or three times a year in one place. There are all enterprise of the industry in this special job fair, therefore there are a lot of jobs for choice.
- Campus recruitment. This is for new graduates, world 500 strong, foreign companies and local companies, can be seen in the campus recruitment.
- Network job. Online recruitment has been deeply into all walks of life, most companies have a recruitment on the Internet. In each big talent net, job hunters can get a lot of post information on the website.

- Recommendation of headhunters. A lot of high-level jobs are hiring workers by headhunters. Generally, they will actively look for you, once you are choused by headhunters, high-paying jobs are not far away from you.
- Introduced by acquaintance. The opportunity of getting a job by acquaintance is high. But we also need to leave dim sum, whether the job introduced by your friends is suitable for you.
- The company straight stroke. Some companies send a few people set up a stall by the side of high pedestrian road, direct hire. Such kind of hiring is compared commonly emergency, so it's easy to get to work.
- To apply for a job initiatively. If you are very interested in some company, but there is no recruitment requirements, you can try to take the initiative to contact the company recruitment, you can make a phone call to ask, or write a attached resume cover letter sent to the company, for a job.
- Recruitment on newspapers and magazines. This is a very traditional employment channels. Before the rise of network recruitment, this is a very important channel.Despite the impact of network recruitment, but is still a good way.

1.3 The shortage of the traditional employment way

Traditional way of hunting job makes getting a job and the way to change your job numerous, but also there are some disadvantages. Such as campus recruitment, as its name implied is to employ new graduates. There are some youth having dream, alive, but have no experience, sometimes called GuanPeiSheng is synonymous with odd jobs. Network job, developed in the network today, the Internet is filled with a lot of false information, job seekers need to remain vigilant. The cost of recruitment on newspapers and magazines , especially the front page or important layout is too expensive for the company; Audience is widely in the channel of conditions of applicants, to some extent increasing the workload and difficulty of selection of human resources. Such as these, the traditional way to apply for a job is inefficient, choice is narrow, the quantity is limited, the quality of talents is very difficult to have a breakthrough, is not conducive to organizational change and innovation, in order to make the advantage in the jobs to apply for a job, we should consider their own knowledge, working age, work attitude, psychological quality and health level, at the same time we should consider by combining with the reality .In the following process of solving the model, we will make a concrete analysis.

II. The Description of the Problem

2.1 The question is how to inspire the transition to the hermit crab?

Hermit crab is one of the most popular pets in the United States, but its abdomen is very soft, in order to protect its belly, they have to depend on other biological shells for protection. The houses of Hermit crab are conch shell, sea shell, the snail shell, even caps for the bad ecological environment. As the hermit crab body grew older, existing shell already cannot satisfy the needs of existing, must find a suitable house, and then, get into shells, hook on the top of the shells with its tail, a few short legs shored up shells inside, long legs stretched out to the outside of the shell, shell with spiny keep mouth. In this way, it is moved into a new home. In addition, shell is also ha good strong environmental protection. However, changing shells is not random, but in the sequence of queue .

2.2 Introduce the way of the hermit crabs molt carapaces and molted model

2.2.1 hermit crabs molt carapaces

When a lonely crab met a beautiful new shell, it immediately checked this shelter, with its feet and tentacles, and run out of the house to try the new residence. If the new shell is a good choice, crab occupy it. It's the classic hermit crab behavior. But if the new shell is too large, crab will not disappointed, hurried out, on the contrary, it will stand beside the new homes waiting for 15 minutes to eight hours. They will not go away until the other crabs appear, each one try the shells. If the shell newcomers also is too big, they will get together, sometimes form a group of 20. However, crabs not gathered in a random arrangement, on the contrary, they are gathered in a line from the biggest to the smallest crab drawing folder to another. When a crab found its new home, in the queue quickly exchanged all crab shells sequentially. The largest crabs in the front line to catch the crabs discarded shells. The second largest crab sojourn to the first big crab shells. And so on. Due to the scarcity shell, hermit crabs finding a new shell is just like people seeking for a good job.

2.2.2 The model of Hermit crabs molt

A hermit crab met a beautiful new shell, it check this shelter immediately with

its feet and tentacles , and then run out of its family to try the size of the new residence. If the new shell is a good choice, crab takes possession of it. They throws the existing shell away, and runs into a new shell. The process of exchanging shells are shown in figure 1 below:



Figure 1: hermit crabs molt in queue

By observing Figure 1, we can see that first NGR1 wanted to change shells, looking for shells outside, suddenly it discovered the shell GZ, NGR1 got on the shell and had a try, but not suitable, so NGR1 wandering around in GZ, after a short while NGR2 came, NGR2 got on and also tried this shell, is not appropriate, then NGR2 also wandering around GZ, such as NGR1 and NGR2 according to size, sorted from big to small, so, continue to increase, and the lines hermit crabs until GR1, GR1 try on GZ is successful, then GR1 GZ, leave GR1 shell GZ1 and left, after GR1 hermit crabs to try GZ1, if appropriate, put on the new shell and leave their shells, leave, not appropriate to wait, until all of the hermit crabs changing new shell.

2.3 Comparison of human and hermit crabs molt carapaces way photograph relative rigidity

It can be spread to the model of people in work by the hermit crabs in shell model. Studying the flow of talent , we should first clear the concept of talent in the organizations. There are many kinds of classification, the use of talents classification is as follows:

Organizational mobility refers to the personnel in the organization between flow in different positions. The contrast diagram between the mode of Hermit crab molt and artificial exchanging:

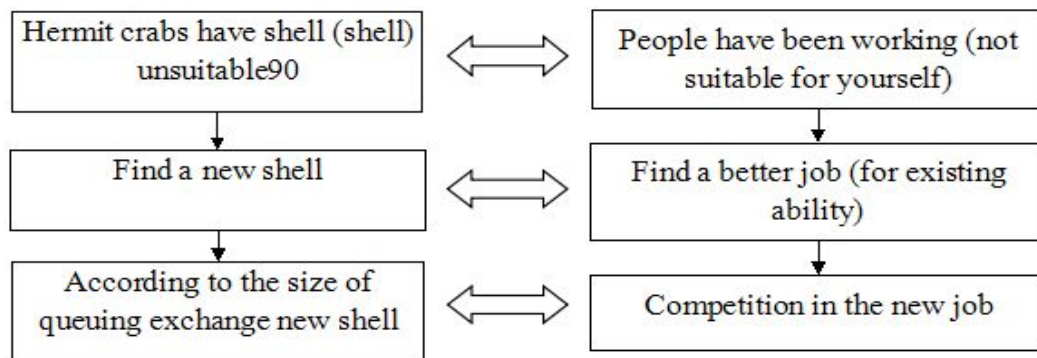


Figure 2: Hermit crabs molt carapaces compared with people in work

Hermit crabs in shell, as the hermit crab grew up, must want to find a suitable for his own house. If you can't find his own house, hermit crab will not be able to better survive. When the hermit crabs found a new house, they will choose to test the size of the house, and then judge the house is not suitable for living, suitable for living, he will go into, if not suitable for living, they will choose to look for a new home, until I find a place to live in the home, then go into.

The ways the human to change jobs, people work increases with time, the growth of the experience, people will be looking forward to more suitable for the development of their work, at this time, people with the existing work, look for other new job, if you find new job is better than your existing job prospects, treatment higher, people will choose the new job at this moment, to an existing job and shove it; If find a job than what you've got, at this time, people will not choose the new job, continue to work in the existing, until one day to find a job that suits their existing ability, people will lose the existing work, to choose the new job.

At this point, you can draw a hermit crab shell with human change, work has a great deal of similarity is as time increases, the capacity increases, the required conditions will improve, so as to find new, for their ability to existing condition, if found in the new conditions cannot meet its current capacity, they will continue to find new conditions, until you find can satisfy the demands of their existing capacity, at this time, they will choose to throw away the existing condition to change the new material conditions.

III. Models

3.1 Symbols

Symbol	Meaning
GZ	Vacancy chain
GR	Flow chain
COV	Crossing-over Value
ACOV	Average Crossing-over Value

CVCO	Coefficient of Variation of Crossing-over Value
VCOV	Variance of Crossing-over Value
CumBenefit	The total benefit from exchange of every work
MeanBenefit	The average benefit from every exchange

3.2 Model assumption and rationality analysis

1. The given information is true and credible;
2. Each job and job seeker can be Quantized by one composite indicator ;
3. Since the creation of jobs and income requirements for this job should be brought in line, so job seekers concerned, he is currently working with the corresponding ability to work on behalf of his current income, namely the interests of job seekers.
4. When a job seeker found a better job, he will change job immediately;
5. Changing jobs does not spend time;
6. There will be some new jobs and job seekers every day, and their quantities submit to some special distribute;
7. We only come down to one professional posts, such as communication engineer, car salesman, etc., in this paper, we only consider office industry;

3.3 The basic principle of the Analytic Hierarchy Process

AHP depending on the nature of the problem and the overall objective to be achieved, the problem is broken down into different constituent elements, and in accordance with inter-related factors and the affiliation between the different levels of aggregation factors combined to form a multi-level analysis of the structural model and ultimately make the problem boils down to the lowest level relative to the top (of the total targets) to determine the relative importance weights or relative merits of the order scheduled (for program decisions, measures, etc.).

1、Analytic Hierarchy Process Steps[2]

- (1) establish a hierarchical model;
- (2) judgment matrix;
- (3) The level of single-sort;
- (4) conformance testing;
- (5) The total level of sort.

In step three of the process needs to be carried out layer by layer.

This article and apply principal component analysis method to classify the condition of the company stratification, and finally in full consultation with the candidates analyzed using hierarchical views on the basis of various aspects of the right method to calculate the relative weight of each index established the following is the ultimate choice for the job of handed class hierarchy diagram, shown in Figure 3:

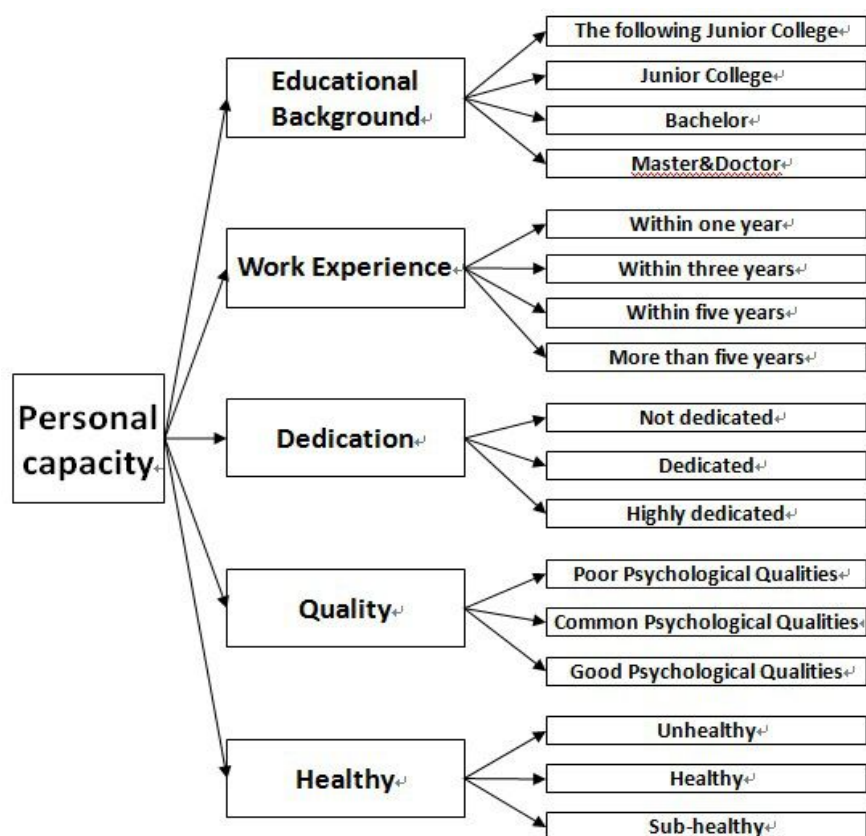


Figure 3: Factors Analytic Hierarchy Figure

2、the establishment of judgment matrix, to determine the index weight

Judgment matrix is more of a factor as the evaluation target layer by layer elements of the pairwise comparison to determine the matrix elements.

3、the establishment of judgment matrix

The first layer or "evaluation system model" evaluation system as the evaluation criteria for the type of talent, academic knowledge, working age, work attitude, psychological, physical pairwise comparisons. Let these five elements are A1, A2, A3, A4, A5 judgment matrix is established as follows:

Table 1: judgment matrix model

Model Evaluation	A1	A2	A3	A4	A5
A1	A11	A12	A13	A14	A15
A2	A21	A22	A23	A24	A25
A3	A31	A32	A33	A34	A35
A4	A41	A42	A43	A44	A45
A5	A51	A52	A53	A54	A55

The measure is judged to represent $A_i A_j$ relative importance on the number of scale, ie the number of forms A_{ij} . Index evaluation model to model, the establishment of judgment scales as defined in Table 2 below:

Table 2: Determine the scale defined

Factor than the factor	Quantized values
Ai and Aj are equally important	1
Ai is slightly more important than Aj	3
Ai stronger important than Aj	5
Ai and Aj highly important	7
Ai and Aj extremely important	9
Two adjacent intermediate value judgments	2, 4, 6, 8

Table 2 is to quantify the value of A_i / A_j 's value, if the calculated ratio A_i / A_j degree of importance, you can by the formula: $A_{ji} = A_j / A_i = 1/A_{ij}$ calculated.

Judgment based metrics, to have corporate recruiters and institutions of these five factors recommended by pairwise comparison of the results to establish the value judgment matrix A, as shown below.

$$\begin{bmatrix} 1 & 2 & 3 & 1 & 2 \\ 1/2 & 1 & 2 & 1/2 & 1/2 \\ 1/3 & 1/2 & 1 & 1/3 & 1/2 \\ 1 & 2 & 3 & 1 & 2 \\ 1/2 & 2 & 2 & 1/2 & 1 \end{bmatrix}$$

4、 Heavy layers of factors to calculate the right

According to AHP theory, we can choose and French roots, characteristic roots and logarithmic least squares method and other methods to determine the weight, relative to the model evaluation method which is good and normal and root method. Selection and method, and the method of this paper is calculated as follows:

$$W_i = \frac{\sum_{j=1}^n A_{ij}}{\sum_{k=1}^n \sum_{j=1}^n A_{kj}}$$

Can be calculated by the formula A1, A2, A3, A4, A5 six factors being right. Specifically as shown in Table 3:

Table 3: six factors form the right weight

Factor	Calculation	Result
A1	WA1=9/ (9+4.5+2.67+9+6)	0.289
A2	WA1=4.5/ (9+4.5+2.67+9+6)	0.144
A3	WA1=2.67/ (9+4.5+2.67+9+6)	0.086
A4	WA1=9/ (9+4.5+2.67+9+6)	0.289
A5	WA1=6/ (9+4.5+2.67+9+6)	0.192

3.4 Basic Model

According to the above analysis, we define the basic model as follows:

3.4.1 Initialization and Parameters setting

Note 1: From[Papers], we know background, work experience, work attitude, psychology and health can be used to measure workers' ability. Their proportionately structures as follows:

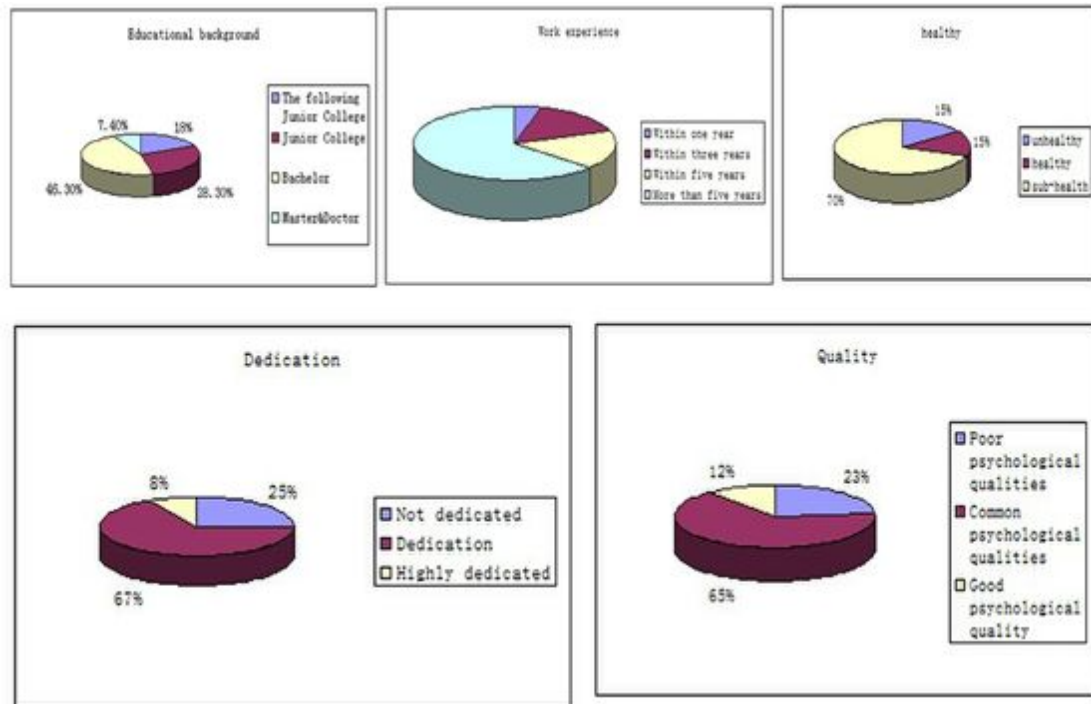


Figure 4: Five kinds of factors influence the proportion of each distribution

When you want to create a job, you just need to create its 5 properties go along with each proportion.

Note 2: When you want to create a job seeker, you should create his job first; then, calculate his job's composite indicator; finally, you could deem that his actual work ability is marginally more than his job's composite indicator.

Note 3: We use AHP (analytic hierarchy process) to calculate composite indicator from 5 element properties:

we calculate the weight of the 5 element properties are (0.289、0.144、0.086、0.289、0.192).

Note 4: We think the jobs as the vacancy chain, the job seekers as the flow chain, the two constitute opportunity chain [3].

Initialization

Step 1: Let education background, work experience, work attitude, psychology and health be element qualities;

Step 2: Randomize jobs in several years(See Note 1), we assume there will be 20 new jobs published every week;

Step 3: Assume there will be 200 new job seekers every week, randomize job seekers in several years (See Note 2), ;

- Step 3: From [papers], we calculate the composite indicator by AHP(See Note 3);
- Step 4: Create matrix J which has 2 columns, the first column denote jobs' composite indicator and the second denote publish date, as new jobs;
- Step 5: Create matrix JA which has 3 columns, the first column denote job seekers' composite indicators and the second their jobs' composite indicators and the third his publish date, as new job seekers;
- Step 6: Let the length of JA be 50 for testing our model, which means the length of the flow chain be 50(See Note 4);

3.4.2 Description of Basic Model

Step 0: Initiation;

Step 1: update the vacancy chain and the flow chain In detail, the first week of the new jobs added to the vacancy chain GZ, new job seekers join the flow chain GR.. The GR according to individual capacity index, which GR first column of data, arranged in a descending order, ie,

$$GR_{1,1} > GR_{2,1} > GR_{3,1} > GR_{4,1} > GR_{5,1} > GR_{6,1}, \dots, > GR_{n,1};$$

These officers have the corresponding ability to work indicators:

$$GR_{1,2}, GR_{2,2}, GR_{3,2}, GR_{4,2}, GR_{5,2}, GR_{6,2}, \dots, GR_{n,2};$$

However, the work ability index and not strictly in the order. Similarly, for GZ are sorted. GZ is less than the minimum operating capacity of GR in part removed. Into Step 2.

Step 2: For the first job GZ, all candidates with existing work to apply for the job, the second column from GR to find the first position is less than the ability to work, denoted by k, then the representative GR in the k-th seekers GR (k) personal capacity, on behalf of GR in the treatment of the k-th current job seekers (see assumption 3), if it means that the k-th seekers capable requirements, and his current income, GR this means that this person is good revenue for a new job currently working income, he is willing to change jobs. So GR (k) will apply for the job, while the existing work to provide it, then, ranked seekers GR (k +1) will be unconditionally inheritance GR (k) work behind him, and so on, behind him seekers will have to re-arrange the work, turn Step3. If GR is smaller than the second column not found ability to work position, turn Step 3.

Step 3 to determine whether all job seekers are finding new jobs, and if so, the end. Otherwise, it will come up a second job, according to Step2 method to choose the right job seekers candidates until all the work was arranged last. Algorithm flowchart

is shown in Figure

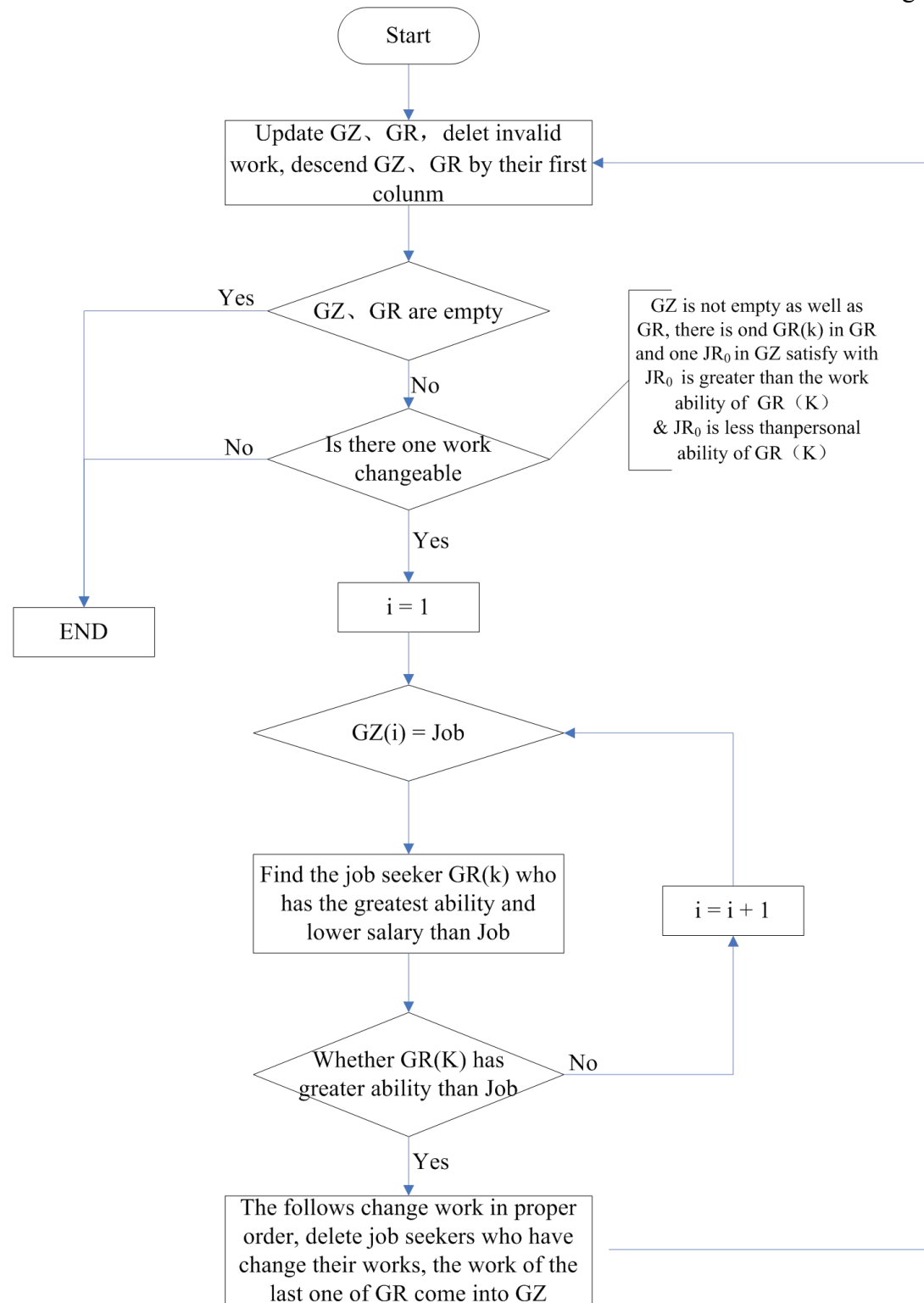


Figure 5: Algorithm flowchart

3.4.3 Problems on the basis of the analysis model

Set the length of time of five years, the chain length of 50 to simulate the situation changes as the number of exchange:

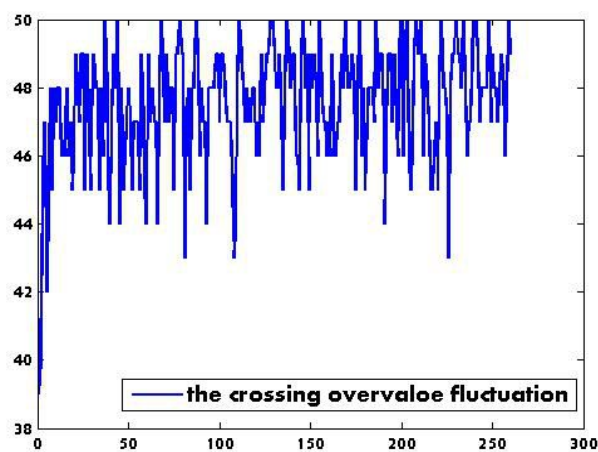


Figure 6: the crossing-overvalue fluctuation

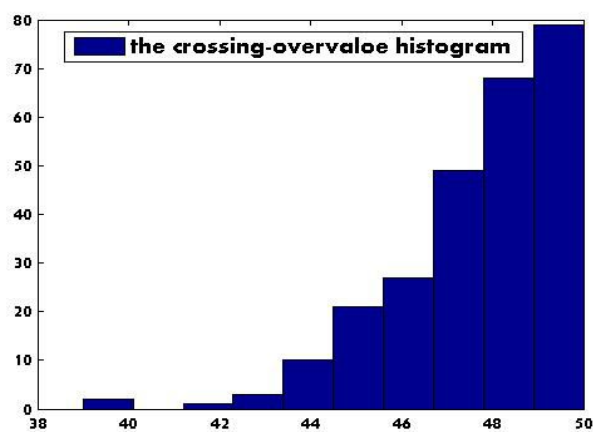


Figure 7: the crossing-overvalue fluctuation

As can be seen from the figure, the number of volatile exchange, the company's human resources management, it is rather difficult, but also caused instability in business operation.

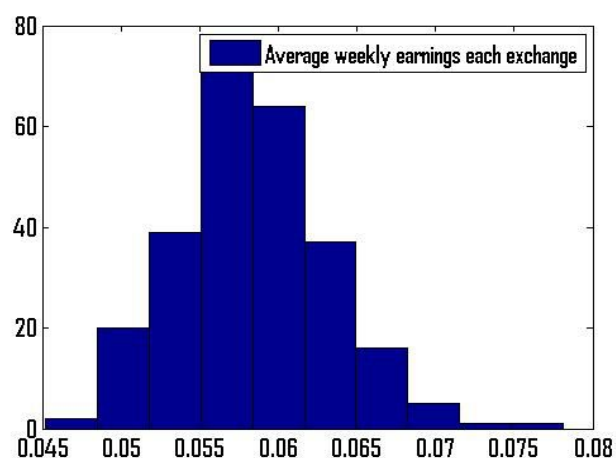


Figure 8 :Average weekly earnings each exchange

As can be seen from the figure that the average exchange revenue growth was normal, for job seekers, this exchange is not fair for job seekers.

Further, in the above flowchart in Fig 1, we can see that this method has the following serious drawbacks:

Assuming the conversion work was done at the moment, it is impossible in real life established;

Assuming flowing exchange of the first person in the chain, followed in turn converted to everyone unconditionally before a person's work. In fact, if the applicant's ability to work back up to the original work of the former than the ability to work, or work behind seekers income exceeds the income of the former, the latter can not successfully change jobs seekers. To overcome these drawbacks, we need to improve the model.

3.5 Improved model

We intend to make are two aspects to this improvement:

3.5.1 Time for work accomplished in collaboration with consumption issues

Collaboration refers to the process of implementation in the target, between departments, coordination and cooperation between the individual and the individual. Collaboration should be multifaceted and broad, as long as the target of a department or a job that must be undertaken to achieve external support and cooperation should be the content of collaboration. Generally include collaborative resources, technology, teamwork, of information.

Goals requires certain resources, including human, financial, material and other aspects. But in the process of collaboration targets, often appear insufficient given sector resources, such as manpower shortage or inadequate equipment. This requires other departments starting from a holistic perspective, give the necessary support of each other, help each other to achieve the common goal of collaboration.

We calculated, which companies can draw jobs that require substitutions, then we can provide these companies made arrangements for those companies who want to change a number of jobs.

Via such collaborative method, the former assume 5 can be revised to: the whole exchange complete in one week.

3.5.2 Solve the basic model of irrationality

The status quo of today's Chinese society , awareness of career planning is still in relatively early stages. This has just begun to spread awareness of career planning to stage university , many students are still in occupation after graduation confused period , at the beginning of occupation on the lost self and career goals , missed the best career development period , not really to put their skills and potential to effectively play. Such a situation, is not conducive to such a large population of Chinese population shift towards power , slowing the process of mining and cultivation of talents in our country , adds a certain extent, the whole society in transition from the low-end manufacturing to high-end manufacturing staff training costs.

The current exchange of people working mainly: he found more suitable than the existing work of the job , and then jump over ; been promoted , rising job . The downside is that the cycle is too long , and there is great uncertainty, is not conducive to the development of their own quality , is not conducive to social development of human resources.

So job agency appears, but its existing technical capability is very old-fashioned, we question the phenomenon inspired by the integration of methods such as title 1) , but this method is too idealistic , can not be used in real life , so on this basis we combine practical , the model was improved .

From Scheme 1 , we found what we should do is to make sure job seekers have enough ability as well as wish of exchanging, that is to say, he has greater composite indicator and the former's work has better salary. So, we revise our model as follows:

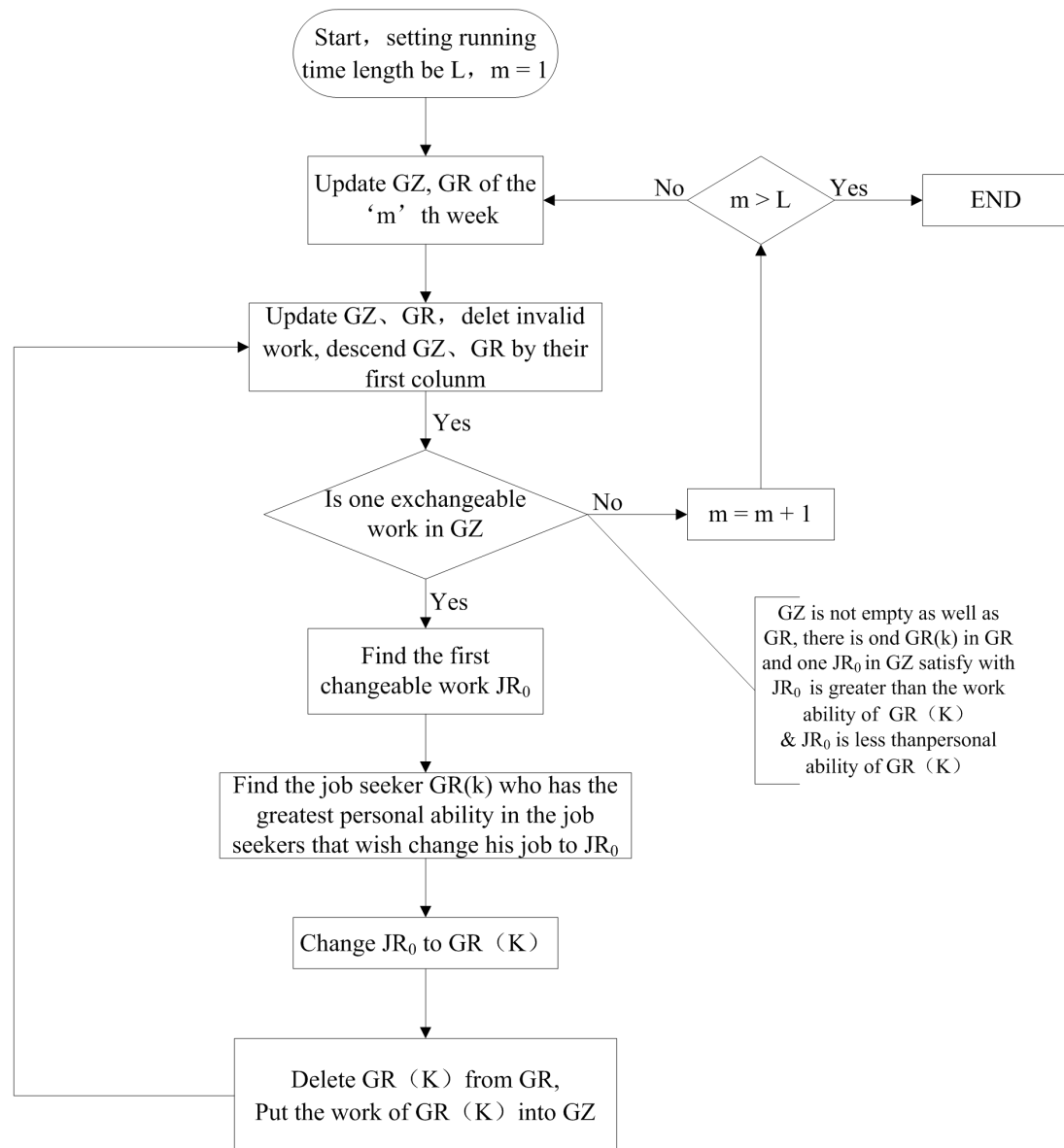


Figure 9 : Model flowchart

3.5.3 Improved model is applicable to which industries

1、candidates choose to work and the way companies work arrangements

After the weight of each factor in the choice of indicators system of heavy work out, you can choose to work to measure the indicators that academic selection factors by five working knowledge of working age, work attitude, psychological, physical fitness index value can be drawn in the queue exchange of social development strategy, so that everyone gets the benefit. And we can draw the following conclusions.

Establish the appropriate model of employment patterns by hermit crabs use mathematical modeling knowledge can ultimately be the most suitable for human pattern of employment.

(2) by a specialized data analysis expert , may be appropriate to set the weight value of each factor , and then follow the principle of AHP, get to work when two or more types of job candidates best choice .

(3) If the data are real and effective , we can promote to other industries , but not all are suitable for this hermit crab industry employment patterns , as some industry employees if circumstances change jobs frequently , can lead to lack of company leadership positions the phenomenon appears difficult to manage operations .

(4) If you want to fit hermit crab pattern of employment , the company must meet the massive change employees will bring more benefits. For example, the sales industry , exchange feasible , teachers and doctors within the same industry, the technology sector.

3.5.4 Model running analysis

Obviously, the longer flow chain provide with the more chances of change jobs, but it will cause change happens too frequently. Our thoughts is define a new indicator to describe the efficiency of change.

Model and the results of the analysis established by this article, you can see the hermit crab shedding patterns useful in cases of human change jobs, for one type of work, with the increase of working-age people, enhance, improve the ability of technology to replace their existing work, the ability to find the most suitable for the present work has considerable significance.

When the chain is set too short, the work is not conducive to people selected, reflecting capacity. If the chain is too long to select the company a lot of substitutions is not conducive to the company's development; therefore select the appropriate chain length, for individuals and companies, have considerable significance.

3.6 Set different chain lengths compared

Know the following instructions:

With COV (i) (Crossing-over Value) represents the first i-week exchange times, COV (i) the average of five years referred to as ACOV (Average Crossing-over Value), its variance VCOV (Variance of Crossing-over Value), with a mean CVCO (Coefficient of income variation of Crossing-over Value) number of representatives of the exchange coefficient of variation, with CumBenefit week on behalf of five years for work seekers obtained through promotion of the total, with representatives of five MeanBenefit the average benefit for each year of work exchange.

In order to found the best length of the flow chain, we set length be 3 : 10: 133, then searching. The result as follows:

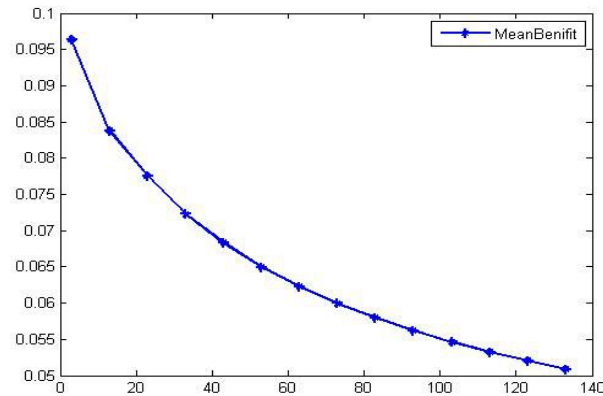


Figure 10: MeanBenefit

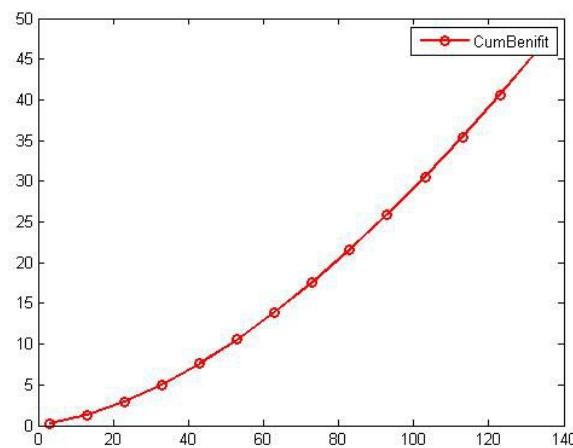


Figure 11: CumBenefit

As can be seen from the figure 10 with increasing chain length of the flow, MeanBenefit showing a decreasing trend, while the figure 6 displays CumBenefit showed an increasing trend, and are approximately in line with a linear relationship. This is because the algorithm priority will be assigned to work best for individuals most capable people, and in the less mobile chain length, and each will have a greater probability of job seekers to get the best jobs. At the same time, growth CumBenefit longer chain length instructions, the more job seekers get a chance to improve the work, which would allow more people to benefit. And because MeanBenefit showed a decreasing trend, which indicates the number of working for the greatly increased, from a business point of view, this will lead to strengthening corporate officers of the degree of instability. We need to find a suitable length, on the one hand allows more rapid employment can enhance revenue, on the other hand enterprise mobility will not be too severe. Thus, we need to find a stable state.

We then analyzed COV (i) the changes.

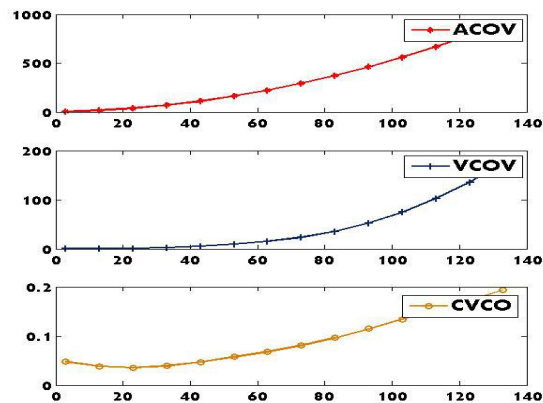


Figure 12:COV CHANGE

With the increase of mobile chain length, ACOV showed an increasing trend, while the shorter time (before 40), VCOV slow increase in the chain length, indicating the degree of exchange discrete number is not strong, relatively stable, while CVCO the chain length [13, 33], there is a local minimum interval, indicating that in this interval, there is a proper chain length makes switching frequency is relatively stable, which means that human resource companies, job seekers can exchange more stable arrangement work, but there are at this time is relatively short chain length, the higher the average job seeker upgrade rate, while companies are relatively stable.

3.7 Select the optimal chain length

We choose the optimum chain length obtained from the above, the analysis can be obtained when the chain length, the average number of people working for the relatively stable.

To this end, we re-interval in accordance with the chain length of 15-35 search step is 2, the following results:

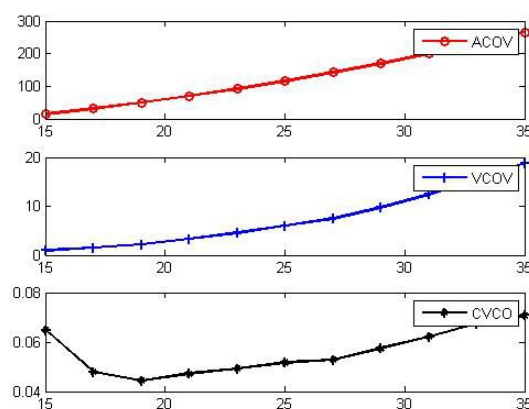


Figure 13:ACOV,VCOV,CVCO

As can be seen from Fig. 8, when the chain length is 19:00, CVCO made minimum. Therefore, we believe the most appropriate chain length is 19. Generally,

human resources companies can choose the length of the flow in the vicinity of 20 to the chain.

3.8 Simulated situation for five years

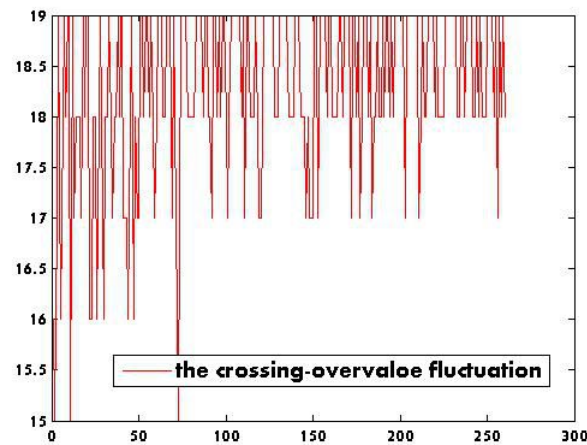


Figure14 :the crossing-overvalue fluctuation

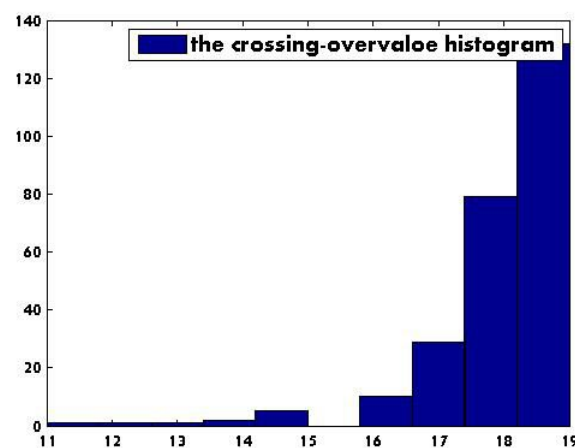


Figure 15:the crossing-overvalue histogram

From the Figure 2 can be seen above, when the flow chain length of 19, the number of daily exchange basically stable at 15-19 times, which means that human resource management from the point of view, as long as the staff set to meet within a week 15-19 the candidate needs to change jobs; from a business perspective it seems, because we set the number of average weekly increase in work for 20, little difference between the two, indicating that such an exchange would not affect the company's normal operation.

In addition, the analysis of fluctuations and fluctuations in the average yield of total revenue, and help to understand the interests of job seekers through the exchange of occupational attainment. When the flow of the chain length is 19, the average exchange each exchange gains and gross weekly earnings volatility as shown below:

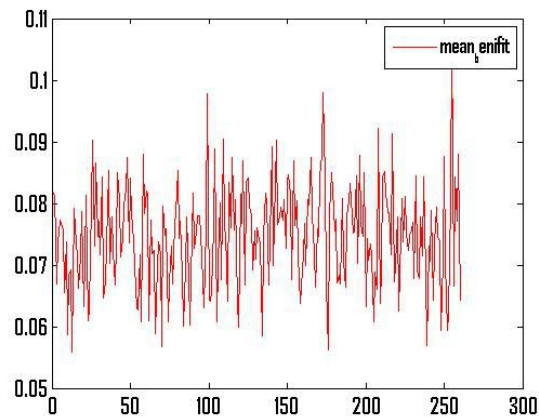


Figure 16: The average income of each exchange

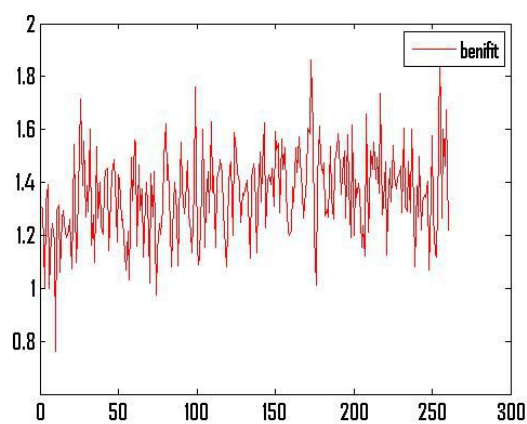


Figure 17 Total revenue weekly exchange

Respective frequency histograms are:

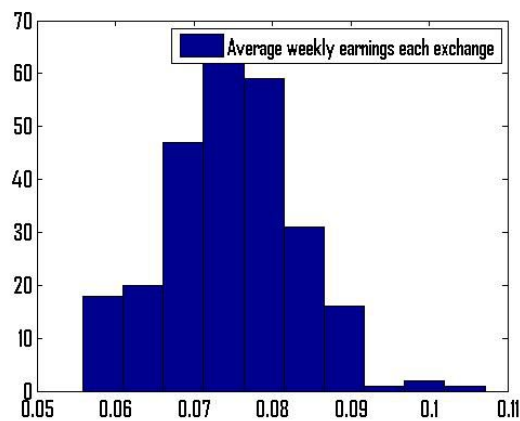


Figure 18: Average weekly earning each exchange

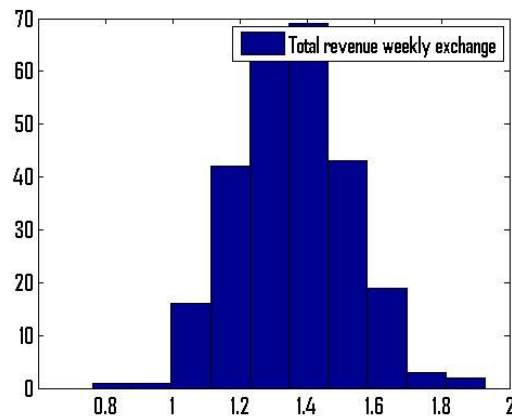


Figure 19: Total revenue weekly exchange

From the above it can be seen in Figure 4, through the exchange of job seekers gain obtained is a relatively high normal, which means that a certain degree of fairness of the exchange upgrade.

3.9 Final results found

Model and the results of the analysis established by this article, you can see the hermit crab shedding patterns useful in cases of human change jobs, for one type of work, with the increase of working-age people, enhance, improve the ability of technology to replace their existing work, the ability to find the most suitable for the present work has considerable significance.

When the chain is set too short, the work is not conducive to people selected, reflecting capacity. If the chain is too long to select the company a lot of substitutions is not conducive to the company's development; therefore select the appropriate chain length, for individuals and companies, have considerable significance. The most suitable chain length is increased by the average weekly new job.

IV. Conclusions

In the current context of economic globalization, the increasingly fierce competition in human resources, corporate human resources management more and more attention. Analysis chance chains will likely become the driving factor in the

talent competition, prompting companies from a strategic height, fully examine the organization's own resources and organizational conditions of the external environment, developed under the organization's vision, organizational goals and strategic planning guidelines organization of future demand and supply planning of human resources to support the implementation of the strategic plan to promote the realization of the vision and goals of the organization.

4.1 Conclusions of the problem

- Hermit crab shedding mode can be used to exchange work.
- Exchange of greatest difficulty lies in the scarcity of jobs and restrictive.

4.2 Methods used in our models

- Calculated using integrated AHP individual ability and job demand capabilities, we assume that job demand capacity equivalent to income from work.
- Advantage of the opportunity chain model based on the construction of the hermit crab shedding mode switching working model.
- By repeating the search frequency of exchange, exchange gains and average per week on average exchange fluctuations and the frequency distribution of the total revenue, results when setting up mobile chain length of the average weekly number of new job nearby, you can achieve the best results.

V. Future Work

5.1 Another model

In order not to reduce corporate profits, leading firms may be out of the market. As the main flow of the enterprise, not simply follow the changes in the market changes, you must change the socio-economic development in the future to predict and forecast consumption trends for the future, and actively create update services, guide consumption. Circulation enterprises in a complex market environment, in order to take advantage of the competition, it must be the development of a global market perspective, grasp the development trend of the global economy, according to the requirements of sustainable development of human society to develop and implement a global scope with the external environment conducive to enterprise resource management strategy match, strive to cultivate core competitiveness of enterprises, in order to ensure the sustainable development of enterprises and sustainable development.

5.1.1 Circulation enterprises manage low-carbon embodied

Circulation enterprises manage carbon, we must recognize that the problem combined circulation enterprises and the market, and the market circulation enterprises production processes and consumption are inseparable, and its circulation itself reflects the market exchange and distribution sectors. Therefore, the management of low-carbon from these four aspects to consider.

First, from the circulation and production processes, the mode of production determines the mode of circulation, and in turn will affect the flow of production. Low-carbon circulation class enterprise management, enterprise class requires the circulation flow of products meet the requirements of the national low-carbon economy, from product raw materials, production technology, production equipment, production environment, must comply with the requirements of a low-carbon economy. Do not meet the requirements of the product, circulation enterprises should adopt different strategies according to the product under its circulation, product requirements and low carbon economy compared to poor circulation enterprises can keep their circulation, or impose excessive fees, thereby enabling manufacturers to re-think the product recovery and processing costs are lower. For products with a low carbon economy requires compliance with the basic product, circulation enterprises may give producers the proper flow of benefits, and for the cost of circulation enterprises, and by state subsidies to support the development of low-carbon circulation enterprises.

Secondly, from the exchange and distribution chain, the product exchange and distribution sectors must have a circulation of products, then the flow of low-carbon business management class, is reflected in the flow of the class of their own low-carbon, less polluting means of transport to take the environment, mode of transport, taking transport packaging can be recycled, such as improving the efficiency of product flow, and in the process, from the circulation enterprises management must focus only on internal management extends to the choice of packaging materials and custom business process extends to products process methods and tools in circulation, circulation and circulation enterprises through the relevant aspects of management, asking them to comply with every aspect of the requirements of low-carbon, and then do the circulation enterprises in low-carbon circulation, so that the flow of enterprise management class the formation of low-carbon atmosphere spread further through the flow of production processes and consumption.

Finally, circulation and consumption sectors, the circulation of the product determines the consumer aspects of the product, so the flow of low-carbon business management classes to enhance the product's low-carbon, which makes consumer products meet the low-carbon consumption requirements, while it also conveys a consumer culture - a low-carbon culture, consumers in the consumer products, product recycling, reuse and other low-carbon requirements will be considered actively classify the product, and then sent to the recycle bin or recycling points, such

as our battery recycling points. Class enterprise management through circulation of enhanced consumption of low-carbon low-carbon, low-carbon consumption and thus conducive to the formation of range.

VI. References

- [1] Zhang Min user attributes 2012-2013 Survey office in China industry;
http://zdc.zol.com.cn/346/3469833_all.html, 2013 年 1 月
- [2] He is the wind, probability and mathematical statistics analysis; Machinery Industry Press, May 2012
- [3] Xu Jie, composition and application opportunities chain in corporate human resources management; Nanjing University of Technology, 2010
- [4] Liyu Xiu Dai Chengqiu, analytic hierarchy process to build students' comprehensive quality evaluation system; History Expo (theory), June 2011

VII. Appendix

Parametersetting:

```
N_Jobperday = 20;
N_days = 52*5;
N_orig_JobApp = 200*N_days;
N_orig_JobApply = N_Jobperday*N_days;
```

Compassess:

```
function [ability] = compassess(origin_abb)
wight = [0.289 0.144 0.086 0.289 0.192];
ind_ori_abi = origin_abb(1:5,:);
ind_abi = wight*ind_ori_abi;
ability = [ind_abi;origin_abb(6,:)];
ability = ability';
```

randomgjob:

```
function [Job] = randomgjob(n,m)
%产生 n 个工作,在 m 天内发布
a = 1:n;
b = randsrc(1,n,[1 2 3 4 5; 0.3 0.28 0.32 0.09 0.01]);
c = randsrc(1,n,[1 2 3 4 ; 0.2 0.3 0.4 0.1]);
d = randsrc(1,n,[1 2 3 ; 0.35 0.45 0.2]);
e = randsrc(1,n,[1 2 3 ; 0.3 0.55 0.15]);
f = randsrc(1,n,[1 2 3 4 ; 0.15 0.3 0.4 0.15]);
g = randi(m,1,n);
Job1 = [b;c;d;e;f;g];
Job = compassess(Job1);
```

Randomgja:

```
function [JA] = randomgja(n,m)
%产生 n 个求职者,在 m 天内求职,
a = 1:n;
b = randsrc(1,n,[1 2 3 4 5; 0.3 0.28 0.32 0.09 0.01]);
c = randsrc(1,n,[1 2 3 4 ; 0.2 0.3 0.4 0.1]);
d = randsrc(1,n,[1 2 3 ; 0.35 0.45 0.2]);
e = randsrc(1,n,[1 2 3 ; 0.3 0.55 0.15]);
f = randsrc(1,n,[1 2 3 4 ; 0.15 0.3 0.4 0.15]);
g = randi(m,1,n);
J = [b;c;d;e;f;g];
PA0_2 = compassess(J);
JA_1 = PA0_2(:,1)+max(0,(1+randn(n,1)/2)/10);
JA = [JA_1 PA0_2];
```

initial_JA_J:

```
Job_ori = randomgjob(N_orig_JobApply,N_days);
Job_ori(:,1) = Job_ori(:,1)+linspace(0.2,1.5,N_orig_JobApply)';
JobAppl_ori = randomgja(N_orig_JobApp,N_days);
Vacancy_chain_0 = randomgjob(N_vachain,1);
Chance_chain_0 = randomgja(N_chchain,1);
```

Currjob:

```
function [cuj] = currjob(x,m)
indx = find(x(:,end)==m);
cuj = x(indx,:);
```

currja:

```
function [cuj] = currja(x,m)
indx = find(x(:,end)==m);
cuj = x(indx,:);
```

personabigr:

```
function [y] = personabigr(x)
lambda = 0.933;
ratio = 0.0905;
temp = x(:,1)*(1+ratio)^(1/365);
y = [temp x(:,2:3)];
```

pasort:

```
function [y] = pasort(x)
y1 = sortrows(x,1);
y1(find(y1(:,1)==0),:)=[];
y1(find(y1(:,end)==0),:)=[];
y = flipud(y1);
```

zuizhong1: Chain=19:10:19

```
clear;close all;clc;
parametersetting;
N_vachain = 20;
```

```

N_chchain = 80;
initial_JA_J;
J = [Vacancy_chain_0];
JA = [Chance_chain_0];
passive_ch = zeros(1,N_days);
active_ch = zeros(1,N_days);
benifit = active_ch;
startchainlength = 19;
gap = 10;
endchainlength = 19;
for kkk = startchainlength:gap:endchainlength
    for m = 1:N_days;
        current_JA = currja(JobAppl_ori,m);
        JA = [JA;current_JA];
        JA = personabigr(JA);
        JA = pasort(JA);
        current_J = currjob(Job_ori,m);
        J = [J;current_J];
        deleteJ = find(J(:,1)<=min(JA(:,2)));
        J(deleteJ,:) = [];
        J = pasort(J);
        mm = min(length(JA(:,1)),kkk);
        temp_ch_ind = randperm(length(JA(:,1)));
        ch_ind = temp_ch_ind(1:mm);
        JA = JA(ch_ind,:);
        JA = pasort(JA);
        hrwaste(m) = sum(JA(:,1)-JA(:,2));
        [len_vachain,nouse] = size(J);
        while isempty(J) == 0 && isempty(JA) == 0
            flag = 0;
            JA = pasort(JA);
            J = pasort(J);
            for i = 1:length(J(:,1))
                allk = find(JA(:,2)<J(i,1));
                if isempty(allk) == 1
                    continue
                end
                if JA(allk(1),1) > J(i,1);
                    flag = 1;
                    continue
                end
            end
            end
            if flag == 0
                break
            end
        end
    end
end

```

```

        end
        if J(1,1) < min(JA(:,2)) || J(end,1) > JA(1,1)
            break
        end
        for i = 1:length(J(:,1))
            allk = find(JA(:,2) < J(i,1));
            if isempty(allk) == 1
                continue;
            end
            k = allk(1);
            if JA(k,1) > J(i,1)
                benifit(m) = benifit(m) + J(i,1) - JA(k,2);
                active_ch(m) = active_ch(m) + 1;
                J(i,2) = 0;

                Ch_time = 3;
                temp1 = JA(k,2);
                temp2 = m; % + randi(Ch_time, 1);
                newJ = [temp1, temp2];
                J = [J; newJ];
                JA(k,1) = 0;
                JA(k,3) = 0;
            else
                J(i,2) = J(i,2) + 1;
            end
        end
    end

    mean_exchange_frequency(kkk) = mean(active_ch);
    var_exchange_frequency(kkk) = var(active_ch);
    mean_benifit{kkk} = benifit./active_ch;
    cum_benifit{kkk} = benifit;
end

end
figure
plot(1:N_days, active_ch)
figure
hist(active_ch)
zuizhong2: Chain=3:10:133
clear; close all; clc;
parametersetting;
N_vachain = 20;
N_chchain = 80;
initial_JA_J;
J = [Vacancy_chain_0];

```

```

JA = [Chance_chain_0];
passtive_ch = zeros(1,N_days);
active_ch = zeros(1,N_days);
benifit = active_ch;
startchainlength = 3;
gap = 10;
endchainlength = 133;
for kkk = startchainlength:gap:endchainlength
    kkk
    for m = 1:N_days;
        current_JA = currja(JobAppl_ori,m);
        JA = [JA;current_JA];
        JA = personabigr(JA);
        JA = pasort(JA);
        current_J = currjob(Job_ori,m);
        J = [J;current_J];
        deleteJ = find(J(:,1)<=min(JA(:,2)));
        J(deleteJ,:) = [];
        J = pasort(J);
        mm = min(length(JA(:,1)),kkk);
        temp_ch_ind = randperm(length(JA(:,1)));
        ch_ind = temp_ch_ind(1:mm);
        JA = JA(ch_ind,:);
        JA = pasort(JA);
        hrwaste(m) = sum(JA(:,1)-JA(:,2));
        [len_vachain,nouse] = size(J);
        while isempty(J) == 0 && isempty(JA) == 0
            flag = 0;
            JA = pasort(JA);
            J = pasort(J);
            for i = 1:length(J(:,1))
                allk = find(JA(:,2)<J(i,1));
                if isempty(allk) == 1
                    continue
                end
                if JA(allk(1),1) > J(i,1);
                    flag = 1;
                    continue
                end
            end
            end
            if flag == 0
                break
            end
        end
    end
end

```

```

        if J(1,1) < min(JA(:,2)) || J(end,1) > JA(1,1)
            break
        end
        for i = 1:length(J(:,1))
            allk = find(JA(:,2) < J(i,1));
            if isempty(allk) == 1
                continue;
            end
            k = allk(1);
            if JA(k,1) > J(i,1)
                benefit(m) = benefit(m) + J(i,1) - JA(k,2);
                active_ch(m) = active_ch(m) + 1;
                J(i,2) = 0;
                Ch_time = 3;
                temp1 = JA(k,2);
                temp2 = m; % + randi(Ch_time, 1);
                newJ = [temp1, temp2];
                J = [J; newJ];
                JA(k,1) = 0;
                JA(k,3) = 0;
            else
                J(i,2) = J(i,2) + 1;
            end
        end
    end

    mean_exchange_frequency(kkk) = mean(active_ch);
    var_exchange_frequency(kkk) = var(active_ch);
    mean_benefit{kkk} = benefit./active_ch;
    cum_benefit{kkk} = benefit;
end

figure
subplot(3,1,1); plot(startchainlength:gap:endchainlength, mean_exchange_frequency(startchainlength:
gap:endchainlength), 'or-');
subplot(3,1,2); plot(startchainlength:gap:endchainlength, var_exchange_frequency(startchainlength:
gap:endchainlength), '+b-');
subplot(3,1,3); plot(startchainlength:gap:endchainlength, var_exchange_frequency(startchainlength:
gap:endchainlength)./mean_exchange_frequency(startchainlength:gap:endchainlength), '*k-');
for chis = startchainlength:gap:endchainlength
    mb((chis-startchainlength)/gap+1) = mean(mean_benefit{chis});
    cb((chis-startchainlength)/gap+1) = mean(cum_benefit{chis});
end

figure
plot(startchainlength:gap:endchainlength, mb, '*-');

```

```

figure
plot(startchainlength:gap:endchainlength,cb,'ro-');
zuizhong3: Chain=15:2:35
clear;close all;clc;
parametersetting;
N_vachain = 20;
N_chchain = 80;
initial_JA_J;
J = [Vacancy_chain_0];
JA = [Chance_chain_0];
passtive_ch = zeros(1,N_days);
active_ch = zeros(1,N_days);
benefit = active_ch;
startchainlength = 15;
gap = 2;
endchainlength = 35;
for kkk =startchainlength:gap:endchainlength
    kkk
    for m = 1:N_days;
        current_JA = currja(JobAppl_ori,m);
        JA = [JA;current_JA];
        JA = personabigr(JA);
        JA = pasort(JA);
        current_J = currjob(Job_ori,m);
        J = [J;current_J];
        deleteJ = find(J(:,1)<=min(JA(:,2)));
        J(deleteJ,:) = [];
        J = pasort(J);
        mm = min(length(JA(:,1)),kkk);
        temp_ch_ind = randperm(length(JA(:,1)));
        ch_ind = temp_ch_ind(1:mm);
        JA = JA(ch_ind,:);
        JA = pasort(JA);
        hrwaste(m) = sum(JA(:,1)-JA(:,2));
        [len_vachain,nouse] = size(J);
        while isempty(J) == 0 && isempty(JA) == 0
            flag = 0;
            JA = pasort(JA);
            J = pasort(J);
            for i = 1:length(J(:,1))
                allk = find(JA(:,2)<J(i,1));
                if isempty(allk) == 1
                    continue
                end
            end
        end
    end
end

```

```

        if JA(allk(1),1) > J(i,1);
            flag = 1;
            continue
        end
    end
    if flag == 0
        break
    end
    if J(1,1) < min(JA(:,2)) || J(end,1) > JA(1,1)
        break
    end
    for i = 1:length(J(:,1))
        allk = find(JA(:,2) < J(i,1));
        if isempty(allk) == 1
            continue;
        end
        k = allk(1);
        if JA(k,1) > J(i,1)
            benifit(m) = benifit(m) + J(i,1) - JA(k,2);
            active_ch(m) = active_ch(m) + 1;
            J(i,2) = 0;
            Ch_time = 3;
            temp1 = JA(k,2);
            temp2 = m; % + randi(Ch_time,1);
            newJ = [temp1, temp2];
            J = [J; newJ];
            JA(k,1) = 0;
            JA(k,3) = 0;
        else
            J(i,2) = J(i,2) + 1;
        end
    end
end

mean_exchange_frequency(kkk) = mean(active_ch);
var_exchange_frequency(kkk) = var(active_ch);
mean_benifit{kkk} = benifit./active_ch;
cum_benifit{kkk} = benifit;
end
end

figure
subplot(3,1,1); plot(startchainlength:gap:endchainlength, mean_exchange_frequency(startchainlength:gap:endchainlength), 'or-');
subplot(3,1,2); plot(startchainlength:gap:endchainlength, var_exchange_frequency(startchainlength:gap:endchainlength), '+b-');

```

```
subplot(3,1,3);plot(startchainlength:gap:endchainlength,var_exchange_frequency(startchainlength:
gap:endchainlength)./mean_exchange_frequency(startchainlength:gap:endchainlength),'*k-');
for chis =startchainlength:gap:endchainlength
    mb((chis-startchainlength)/gap+1) = mean(mean_benefit{chis});
    cb((chis-startchainlength)/gap+1) = mean(cum_benefit{chis});
end
figure
plot(startchainlength:gap:endchainlength,mb,'*-');
figure
plot(startchainlength:gap:endchainlength,cb,'ro-');
```